IT Engineer

CASE PORTFOLIO: IT ENGINEERING

2008 ~ 2026

1. GOVERNANCE AND METHODOLOGICAL FRAMEWORKS

CASE #1 Critical Technical Debt Reduction

Large Marketplace

CONTEXT (CHALLENGE)

Mapping and prioritization of refactoring based on risk and change frequency.

SITUATION

The maintenance cost of legacy modules consumed 70% of the development capacity.

TASK

The decision was to prioritize the technical debts that caused the most incidents in production.

ACTION

I used cyclomatic complexity metrics and bug volume to define the refactoring backlog.

RESULT

Freed up 30% of the team's capacity for new features.

CASE #2 Regression Testing Optimization

Enterprise Software

CONTEXT (CHALLENGE)

Automation of the testing pyramid to guarantee quality without delaying the release.

SITUATION

Manual QA cycles took 1 week, preventing daily deploys.

TASK

The decision was to invest heavily in automated integration and unit tests.

ACTION

I implemented parallel test execution in the pipeline, reducing feedback time for developers.

RESULT

Achieved 80% automated test coverage, enabling 5 confident deploys per day.

2. INFRASTRUCTURE AND CLOUD MODERNIZATION

CASE #3 **Cloud Workload Migration**

Retail Tech / Cloud Ops

CONTEXT (CHALLENGE)

Migration of critical workloads to AWS/Azure using an Infrastructure as Code (Terraform) approach. Demonstrates mastery in scalable architecture and migration of on-premises environments to the cloud.

SITUATION

The maintenance cost of physical servers was high, and scalability during peak dates (Black Friday) was insufficient.

TASK

The decision was to migrate 100% of the core application to the cloud in an automated and secure manner.

ACTION

I designed the landing zone on AWS, used Terraform to provision resources, and implemented Auto Scaling and Load Balancers.

RESULT

40% reduction in fixed costs and 99.99% availability during traffic peaks.

CASE #4 **CI/CD Pipeline Automation**

Fintech / Devsecops

CONTEXT (CHALLENGE)

Implementation of automated pipelines using Jenkins/GitHub Actions and Docker. Focus on the efficiency of the development lifecycle and the reduction of manual errors.

SITUATION

The deploy process was manual, took hours, and frequently caused downtime due to human error.

TASK

The decision was to create a continuous integration and delivery flow with automated validations.

ACTION

I configured CI/CD pipelines with unit tests, static code analysis (SonarQube), and automated deployment to Kubernetes.

RESULT

Reduced deployment time from 4 hours to 15 minutes and dropped rollbacks by 70%.

CASE #5

Disaster Recovery & Security Protocols

Healthcare / Digital Hospital

CONTEXT (CHALLENGE)

Implementation of security protocols, encryption at rest, and Disaster Recovery strategies. Demonstrates the ability to protect critical assets and ensure business continuity.

SITUATION

The company was vulnerable to ransomware attacks and lacked a tested disaster recovery plan.

TASK

The decision was to shield the internal network and establish a rigorous backup and redundancy policy.

ACTION

I implemented next-generation firewalls, MFA authentication, and a geo-redundant disaster recovery site with a 15-minute RPO.

RESULT

Passed the ISO 27001 security audit and successfully mitigated two external intrusion attempts.

3. PEOPLE MANAGEMENT AND TEAM DYNAMICS

CASE #6

Knowledge Sharing and Tech Talks

It Consulting

CONTEXT (CHALLENGE)

Creation of architecture guides and conducting Tech Talks to disseminate best practices. Focus on how technical knowledge is shared to raise the maturity level of the team.

SITUATION

There was a large technical disparity among team members, resulting in heterogeneous code quality.

TASK

The decision was to establish a technical mentoring program and shared architecture standards.

ACTION

I instituted weekly Pair Programming sessions and created a living documentation (Wiki) of coding standards.

RESULT

Increased the perceived seniority of the team and reduced code review time by 30%.

CASE #7 ## Production Incident War Rooms

Global E-Commerce

CONTEXT (CHALLENGE)

Leadership in War Rooms for critical incident resolution in production environments. Ability to act under pressure to restore services and identify root causes.

SITUATION

A critical failure in the payment gateway halted sales during a promotional event.

TASK

The decision was to lead the technical task force to identify and fix the bottleneck in record time.

ACTION

I coordinated real-time log analysis, identified a connection pool exhaustion in the database, and applied an emergency patch.

RESULT

Service restored in less than 30 minutes and a contingency plan created to prevent recurrence.

CASE #8 ## Career Development Tracks (IDP)

Saas Enterprise

CONTEXT (CHALLENGE)

Structuring growth tracks for engineers (Y-shape: Specialist or Manager).

SITUATION

Senior engineers were leaving the company because they felt they had 'hit a ceiling' technically.

TASK

The decision was to create a technical career track parallel to management, valuing deep expertise.

ACTION

I defined the responsibilities for Staff and Principal Engineer roles, establishing clear promotion criteria.

RESULT

Reduced technical turnover (churn) by 15% after implementing the technical track.

4. GOVERNANCE AND OPERATIONAL EFFICIENCY

CASE #9

Cloud Cost Optimization (FinOps)

Tech Retail

CONTEXT (CHALLENGE)

Financial management of cloud resources to reduce waste without affecting performance.

SITUATION

The monthly cloud invoice was increasing by 15% month-over-month without a proportional increase in traffic.

TASK

The decision was to implement spending control policies and shut down non-productive resources.

ACTION

I mapped idle resources, implemented reserved instances, and automated schedule-based scaling.

RESULT

Immediate 22% reduction in the monthly invoice and greater budget predictability.

CASE #10

Multi-Cloud and Redundancy Strategy

Logistics Sector

CONTEXT (CHALLENGE)

Implementation of a distributed architecture across different providers (AWS/GCP).

SITUATION

Reliance on a single provider created business risk in the event of major regional outages.

TASK

The decision was to design a cloud-agnostic architecture to guarantee resilience.

ACTION

I configured traffic routing via Cloudflare for failover between clouds and multi-cluster Kubernetes orchestration.

RESULT

99.99% availability achieved with automatic failover between cloud providers.

CASE #11 Systems Audit and IPO Readiness

Fintech Growth

CONTEXT (CHALLENGE)

Ensuring all IT processes comply with rigorous standards (SOX/LGPD).

SITUATION

The company needed to go public, but access and log management were informal.

TASK

The decision was to formalize the change management process and audit trail.

ACTION

I implemented identity management tools and automated log collection for auditing purposes.

RESULT

Full compliance achieved in 8 months, allowing the IPO to proceed without caveats.

CASE #12 Application FinOps for Engineers

Streaming Service

CONTEXT (CHALLENGE)

Training the dev team to write optimized code that consumes fewer cloud resources.

SITUATION

Serverless function processing costs were spiraling out of control due to inefficient loops.

TASK

The decision was to integrate cost metrics directly into the developer dashboard.

ACTION

I optimized the runtime of the functions and taught memory best practices in ephemeral architectures.

RESULT

35% reduction in Lambda function execution costs through code optimization.

CASE #13 Vendor Selection and Evaluation (RFP)

Multinational Retail

CONTEXT (CHALLENGE)

Technical leadership of the process for choosing new technology partners.

SITUATION

The company was purchasing software that did not integrate with the current ecosystem, creating silos.

TASK

The decision was to create a rigorous technical checklist for evaluating new vendors.

ACTION

I conducted technical Proof of Concepts (PoC) with the finalists to validate real-world scalability.

RESULT

Selected an ERP partner that reduced integration costs by 200k annually.

5. DATA ARCHITECTURE AND BACKEND

CASE #14 **High-Volume Database Migration**

Local Fintech

CONTEXT (CHALLENGE)

Migration of relational bases to managed solutions (RDS/Aurora) with zero downtime.

SITUATION

The on-premises database suffered from I/O bottlenecks and data corruption risks.

TASK

The decision was to migrate to a high-performance managed environment.

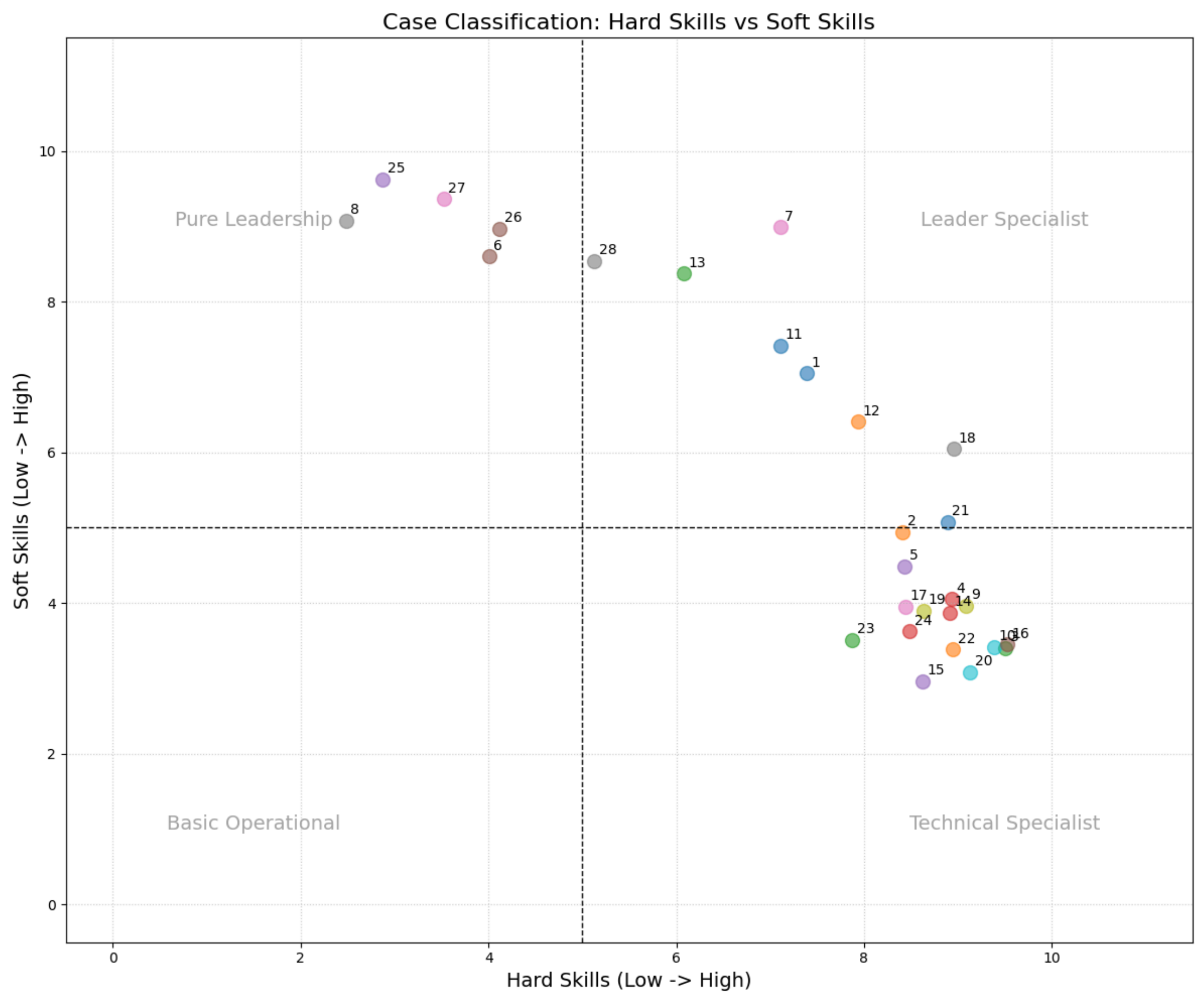
ACTION

I used AWS DMS for continuous replication and performed the switch-over during a low-load window.

RESULT

Migration completed with only 5 seconds of switch-over and a 50% gain in query speed.

CASE PORTFOLIO ANALYSIS: PROJECT MANAGEMENT



Visualization of the distribution of the 28 cases based on the balance between **Hard Skills** (Technical/Processes) and **Soft Skills** (Leadership/Resilience).

CASE #15

Service Mesh Implementation (Istio)

Delivery App

CONTEXT (CHALLENGE)

Managing communication between microservices for enhanced observability and security.

SITUATION

Difficulty in tracking failures across a network of 200 microservices delayed incident resolution.

TASK

The decision was to adopt a service mesh to centralize control and metrics.

ACTION

I implemented Istio in the Kubernetes cluster, configuring Kiali for traffic visualization and telemetry.

RESULT

Total visibility of east-west traffic and implementation of mTLS across the entire service mesh.

CASE #16

Real-Time ETL Pipeline (Kafka)

Industry 4.0

CONTEXT (CHALLENGE)

Processing streaming events for immediate decision-making dashboards.

SITUATION

The sales dashboard had a 24-hour delay, preventing quick reactions to stock issues.

TASK

The decision was to replace batch processing with event-driven processing.

ACTION

I implemented Apache Kafka for CDC capture and Spark Streaming for processing.

RESULT

Data available to the business in less than 10 seconds after the transaction.

CASE #17 API Gateway Implementation

Integration Hub

CONTEXT (CHALLENGE)

Standardization, throttling, and security for APIs exposed to partners utilizing Kong/Apigee.

SITUATION

Each API had a different authentication method, making external partner integration difficult.

TASK

The decision was to centralize API management under a single facade.

ACTION

I configured the Kong gateway to handle authentication via JWT and request rate limiting per client.

RESULT

100% of APIs standardized under OAuth2 alongside efficient traffic control.

CASE #18 Legacy Modernization (Strangler Pattern)

Insurance Company

CONTEXT (CHALLENGE)

Gradual replacement of a monolith with microservices without disrupting operations.

SITUATION

The legacy system was impossible to update, yet too critical to shut down all at once.

TASK

The decision was to gradually decouple functionalities using a reverse proxy.

ACTION

I redirected specific calls to new microservices while keeping the old core running.

RESULT

Migrated 40% of the most critical functionalities to microservices within 1 year, with zero downtime.

6. SECURITY AND TECHNICAL COMPLIANCE

CASE #19 Compliance Automation (Policy as Code)

Banking Institution

CONTEXT (CHALLENGE)

Using tools like OPA (Open Policy Agent) to ensure security guardrails in the CI/CD pipeline.

SITUATION

Developers were creating resources outside of company security standards due to a lack of automated guardrails.

TASK

The decision was to automate compliance validation directly in pull requests.

ACTION

I developed governance rules that block the creation of public S3 buckets or open security groups.

RESULT

Automatic blocking of 100% of deployments that did not comply with pre-established security policies.

CASE #20 Secrets and Credentials Management

Payments Fintech

CONTEXT (CHALLENGE)

Centralization and automatic rotation of keys and passwords using HashiCorp Vault.

SITUATION

Database passwords were exposed in configuration files and source code.

TASK

The decision was to use a centralized and dynamic password vault.

ACTION

I integrated applications with HashiCorp Vault, removing hardcoded credentials from files and environment variables.

RESULT

Elimination of plaintext secrets and implementation of automatic rotation every 30 days.

7. PLATFORM ENGINEERING AND SRE

CASE #21

Internal Developer Platform Creation (IDP)

Tech Unicorn

CONTEXT (CHALLENGE)

Development of a portal for developers to perform self-service infrastructure provisioning (Backstage).

SITUATION

The infrastructure team was a bottleneck, taking days to deliver a new database or queue.

TASK

The decision was to create an abstraction layer to give developers autonomy.

ACTION

I implemented the Backstage portal with pre-approved infrastructure templates via Terraform.

RESULT

Reduced provisioning Lead Time from 3 days to 10 minutes via self-service.

CASE #22

Monitoring and Observability (L.E.T.S.)

E-Commerce

CONTEXT (CHALLENGE)

Implementation of Golden Signals using Prometheus and Grafana.

SITUATION

The team only discovered issues through client complaints, lacking proactive alerts.

TASK

The decision was to establish health metrics (Latency, Errors, Traffic, and Saturation).

ACTION

I configured intuitive dashboards and anomaly alerts based on standard deviation.

RESULT

Incident detection 15 minutes before impacting the end user through intelligent alerting.

CASE #23

Docker Image Standardization (Hardening)

Software House

CONTEXT (CHALLENGE)

Creation of secure and optimized base images for the entire company.

SITUATION

Container images were oversized (GBs) and contained critical vulnerabilities from outdated libraries.

TASK

The decision was to create a catalog of pre-scanned 'gold' images.

ACTION

I reduced image layers, used distroless bases, and implemented vulnerability scanning in the registry.

RESULT

60% reduction in image size and elimination of 'Critical/High' vulnerabilities.

CASE #24 Load Testing Automation (K6/JMeter)

Ticketing Platform

CONTEXT (CHALLENGE)

Execution of stress tests in a staging environment to predict system limits.

SITUATION

The system systematically crashed during large-scale marketing campaigns.

TASK

The decision was to simulate expected traffic to identify breaking points.

ACTION

I developed scripts in K6 simulating concurrent users and analyzed pod behavior in K8s.

RESULT

Identified memory bottlenecks prior to production, supporting 3x more concurrent users.

CASE #25 Blameless Post-Mortem Culture

Global Tech Co

CONTEXT (CHALLENGE)

Institution of failure analysis processes focused on systemic learning.

SITUATION

Errors were punished individually, which led people to hide technical failures.

TASK

The decision was to focus on 'how the system failed' instead of 'who made a mistake'.

ACTION

I led public and documented failure analysis sessions, focusing on process improvement.

RESULT

Increased transparency and reduced the recurrence of identical incidents by 50%.

CASE #26

Stakeholder Management

Investment Banks

CONTEXT (CHALLENGE)

Translating technological risks into financial impacts for both technical and non-technical boards of directors.

SITUATION

The board did not approve investments in technical debt because they did not understand obsolescence risks.

TASK

The decision was to create a risk matrix associating technical debt with potential revenue losses.

ACTION

I presented executive reports correlating downtime with cost per minute.

RESULT

Budget approval for core refactoring after demonstrating the real risk of financial loss.

CASE #27

Conflict Resolution in Mergers (M&A)

Education Group

CONTEXT (CHALLENGE)

Unifying engineering cultures and tech stacks following an acquisition.

SITUATION

Two engineering teams used opposing tools and competed with each other instead of collaborating.

TASK

The decision was to form a hybrid technical committee to choose the best common stack.

ACTION

I promoted integration workshops and defined objective criteria for tool unification.

RESULT

Successful team integration within 6 months, with 90% retention of key talent.

CASE #28

Innovation Facilitation (Internal Hackathons)

Media Group

CONTEXT (CHALLENGE)

Creating spaces for experimentation with new technologies (Generative AI, etc.).

SITUATION

The team was 100% focused on maintenance, with no time to test technologies that could provide a competitive edge.

TASK

The decision was to reserve 10% of sprint time for research and innovation projects.

ACTION

I organized quarterly hackathons with prizes for the best innovative technical solutions.

RESULT

Emergence of 3 new AI-based product features derived from internal prototypes.